

Nakuja Project

Solid Propulsion Team

N4 One-Grain Static Test — 60:40 KNO₃:Sorbitol Formulation

Document Version: 1.0 (Post-Execution Report)
Motor Configuration: N4, 1-Grain, Aluminium Casing, BATES Geometry
Propellant Ratio: 60% KNO₃ : 40% Sorbitol (previously 65:35)
Date of Execution: Friday, 17 July 2026 (Afternoon)
Prepared by: Paschal Peters (Data Recorder / Timer)
Outcome: **SUCCESS** — Full burn achieved; combustion irregularities noted

1 Purpose

This report documents the static fire test of the N4 one-grain solid rocket motor conducted on the afternoon of Friday, 17 July 2026 — the second successful static test in as many days, following the 16 July 2026 test of the 65:35 KNO₃:Sorbitol grain. For this test, the propellant formulation was changed to a **60:40 KNO₃:Sorbitol** ratio (oxidizer:fuel). While the motor fired successfully and produced a full burn, the test surfaced combustion-quality concerns — an audibly uneven, "wavy" burn, nozzle choking observed on video footage, and black smoke consistent with incomplete combustion — that were not present, or not documented, in the prior test. This report presents the experimental thrust data, compares it against both the unchanged theoretical (OpenMotor) curve and the previous 65:35 experimental run, and records the team's discussion and competing explanations for the observed behaviour.

2 Test Overview

Motor	N4, 1-Grain, Aluminium Casing
Propellant Ratio (this test)	60% KNO ₃ : 40% Sorbitol
Propellant Ratio (previous test, 16 July)	65% KNO ₃ : 35% Sorbitol
Ignition/Procedure Faults	None — successful ignition and full burn
Combustion Behaviour Observed	Audibly uneven / "wavy" burn; nozzle choking visible on video footage; black smoke
Outcome	Success — motor fired fully, but combustion quality concerns raised

3 Theoretical Performance (Unchanged)

The OpenMotor simulation used for comparison is the same model referenced in the 16 July report, as the grain geometry (BATES, 86 mm casing, 145 mm length, 35 mm core, 23 mm throat) was not altered for this test — only the propellant oxidizer:fuel ratio changed. The simulated curve therefore represents the same theoretical target used previously:

Parameter	Simulated Value
Peak Thrust	377.64 N
Time of Peak Thrust	1.38 s
Total Impulse	1014.11 Ns
Burn Time	2.85 s

4 Experimental Results — 60:40 Grain

Parameter	Experimental Value
Peak Thrust	234.18 N
Time of Peak Thrust	3.12 s
Total Impulse	823.74 Ns
Burn Time	8.08 s

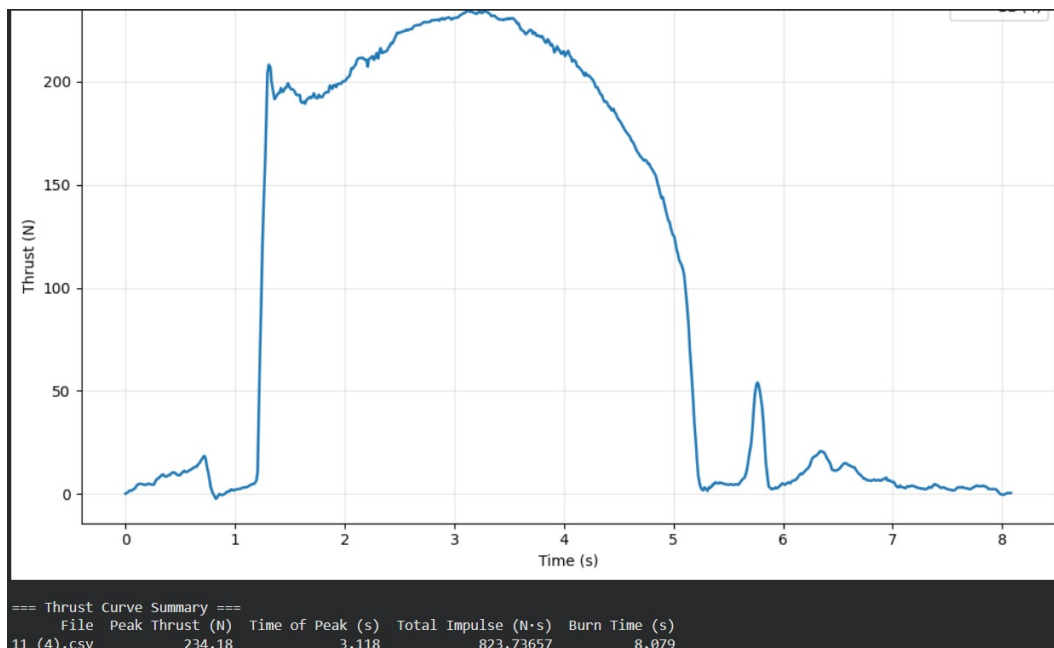


Figure 1: Experimental thrust curve for the 60:40 KNO_3 :Sorbitol grain (17 July, afternoon), with summary statistics.

The curve rises to an initial local peak near 210 N by ~ 1.3 s, dips briefly, then climbs through a broader hump peaking at 234.18 N around 3.12 s, before decaying over a longer tail to burnout at 8.08 s. A distinct

secondary spike (~55 N) appears around 5.7 s, after the main decay, which does not appear in the 16 July curve.

5 Comparison: 60:40 vs. 65:35 vs. Theoretical

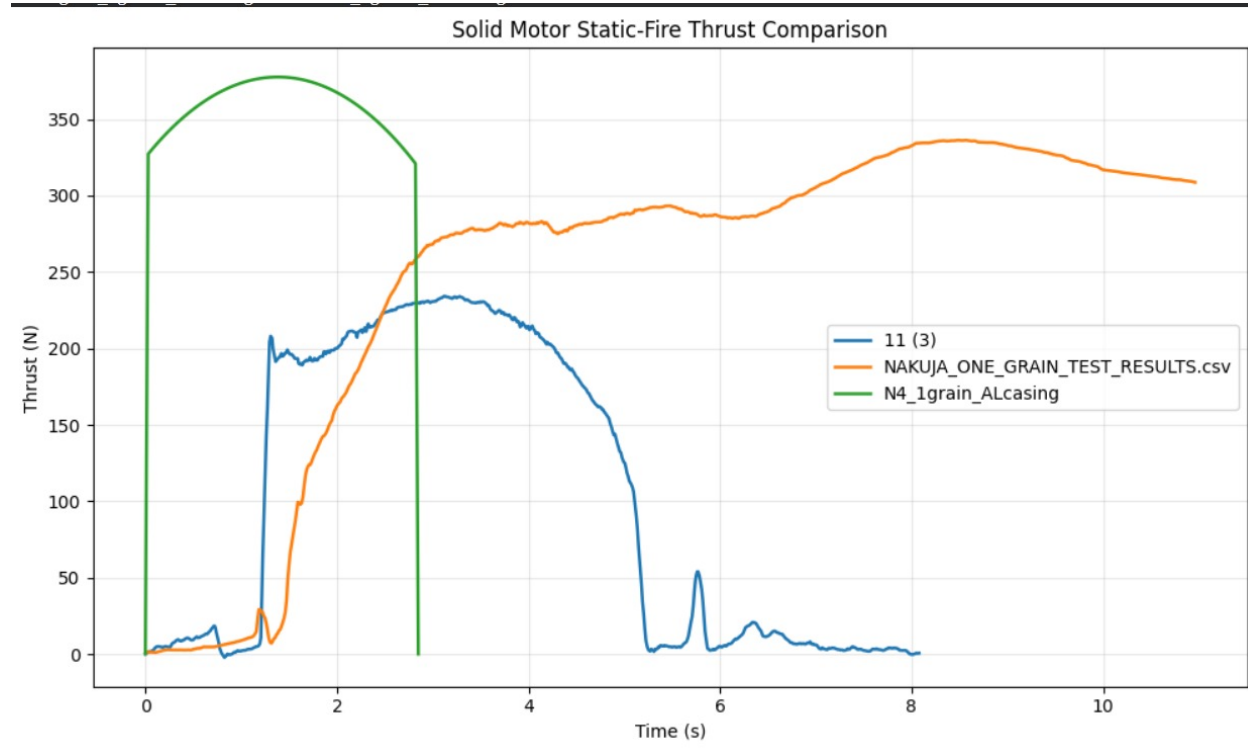


Figure 2: Full thrust curve comparison: 60:40 grain (blue, "11 (3)"), 65:35 grain from 16 July (orange), and OpenMotor simulation (green).

Parameter	Theoretical	65:35 (16 Jul)	60:40 (17 Jul)
Peak Thrust (N)	377.64	336.33	234.18
Time of Peak (s)	1.38	8.49	3.12
Total Impulse (Ns)	1014.11	2719.01	823.74
Burn Time (s)	2.85	10.96	8.08

Relative to the 65:35 grain, the 60:40 formulation produced a **lower** peak thrust (−30%), a much **shorter** burn time (−26%), and substantially **lower** total impulse (−70% relative to 65:35, and also below the theoretical prediction). It reached its peak earlier (3.12 s vs. 8.49 s) but still far later than the simulation's 1.38 s. Combustion, however, was reported as less clean, with audible unevenness, visible nozzle choking, and black smoke — none of which were reported for the 16 July test.

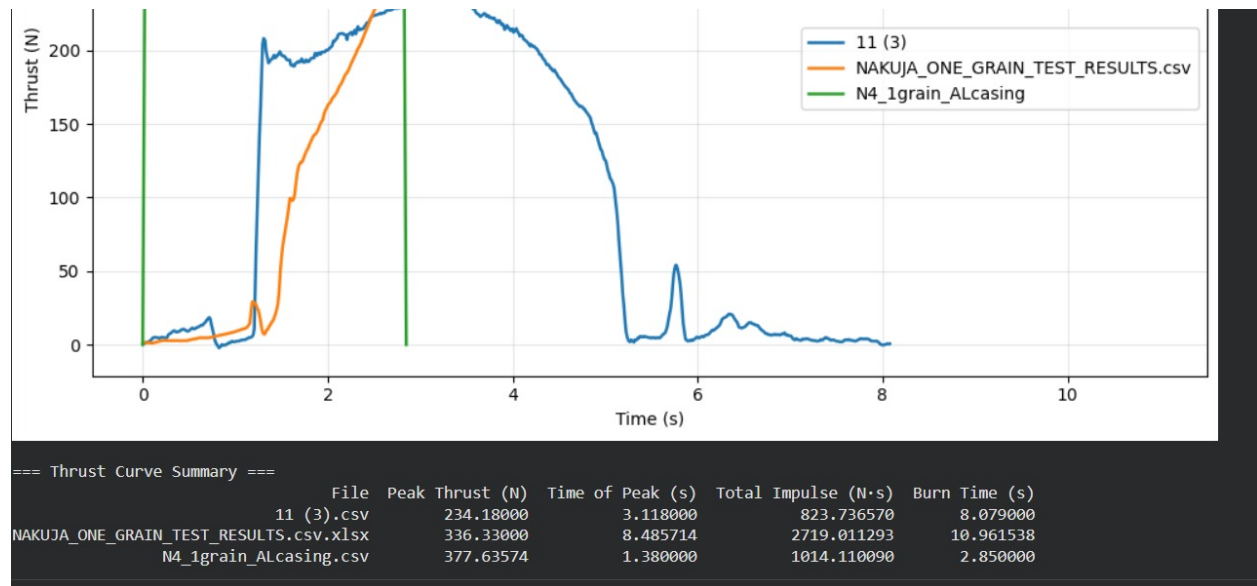


Figure 3: Zoomed view of the ignition and rise region, showing the 60:40 grain's earlier, sharper initial peak relative to the 65:35 grain and the simulation.

6 Observation: Discrepancy Between Audible Burn and Thrust Curve

It was noted during review that although the burn was audibly "wavy" or uneven, and video footage showed visible nozzle choking, the recorded thrust curve does not show corresponding sharp, rapid fluctuations — it appears comparatively smooth aside from the secondary spike near 5.7 s.

This is a genuine and worth-flagging discrepancy rather than an artifact to dismiss. Two contributing factors are worth investigating in parallel with the load cell sensitivity check already proposed:

- **Sampling/response rate:** if the load cell's sampling rate or the logging system's update interval is slower than the timescale of the choking events (which may be sub-second, driven by rapid, transient nozzle blockage and clearing), fast oscillations could be averaged out in the recorded signal even though they are audible and visible in real time.
- **Mechanical/structural damping:** the load cell mount, test stand, or motor-to-sensor coupling may mechanically filter or damp high-frequency force fluctuations before they reach the sensor, so the sensor reports a smoothed net thrust even when the underlying combustion and nozzle flow are locally unsteady.
- **Sensitivity/resolution:** a load cell with a full-scale range set for the expected several-hundred-newton thrust may lack the resolution to clearly register smaller, rapid perturbations superimposed on that baseline — consistent with the sensitivity concern already raised.

Confirming which of these applies would require comparing the load cell's logging rate against an independent time reference from the video footage (e.g., timestamping the audible "wave" events and checking whether the thrust log has data points dense enough, in that same window, to resolve them at all).

7 Team Discussion

7.1 Concern Raised — Dr. Irungu

Dr. Irungu raised a concern regarding whether the grain used in this test had been **properly dried** prior to firing, as a potential explanation for the uneven combustion behaviour observed.

7.2 Response — Bencollins

In response, Bencollins offered the following points, arguing that the grain was well dried and that the observed behaviour is better explained by the change in propellant ratio and its effect on the nozzle:

- The grain being well-dried is evidenced by the **shorter burn time** observed in this test relative to the previous one — a wetter grain would be expected to burn more slowly, not faster.
- This test used a different fuel-to-oxidizer ratio (60:40 KNO₃:Sorbitol) compared to the previous test's 65:35 ratio.
- The uneven burning sound is attributed to **nozzle choking**, which can result either from using a different-sized igniter in this test, or from the change in oxidizer-to-fuel ratio itself introducing a nozzle-clogging problem — consistent with what was observed.
- The **black smoke** observed is attributed to incomplete combustion, as there is insufficient oxygen in the richer (higher-fuel-fraction) mixture to fully consume the carbon present, a known characteristic of KNO₃/sugar-fuel propellants pushed toward a more fuel-rich ratio.

7.3 Outstanding Action Items from Discussion

- **Load cell sensitivity** to be reviewed, both as a possible explanation for the smooth-looking thrust curve despite audible/visual unevenness, and as a general instrumentation check.
- **Grain dryness testing methodology** to be established as a formal, repeatable procedure, so that dryness can be verified objectively rather than inferred indirectly (e.g., from burn time) in future post-test discussions.

8 Conclusion and Next Steps

The 17 July (afternoon) static test of the 60:40 KNO₃:Sorbitol grain is classified as a **SUCCESS** in terms of ignition and full burn completion, with no procedural or hardware faults. However, it delivered substantially lower performance (peak thrust, impulse, and burn time) than both the theoretical prediction and the previous 65:35 grain, and raised new combustion-quality concerns — audible unevenness, nozzle choking, and black smoke — that were not present in the prior test.

- Investigate the thrust-curve/audio discrepancy by cross-referencing load cell sampling rate against video-timestamped choking events.

- Review and, if necessary, recalibrate or replace the load cell for improved sensitivity to short-duration thrust fluctuations.
- Develop and document a standard grain dryness test method, to resolve the dryness question objectively in future reviews.
- Re-run a controlled comparison between 65:35 and 60:40 ratios using **identical** igniter sizing, to isolate whether the nozzle choking is driven by the propellant ratio change, the igniter change, or both.
- Inspect the nozzle post-test for physical residue/clogging consistent with incomplete combustion, to corroborate the black-smoke/oxygen-deficiency explanation.

Nakuja Project — Solid Propulsion Team, KNSB Solid Rocket Motor Development